Predicting and Prioritizing Energy Inefficiency in NYC Buildings

ISYE 4600: Methods & Applications in Machine Learning

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I. Problem Statement and Goal

Energy efficiency in urban environments is a critical and increasingly important challenge, particularly in densely populated cities such as New York City. Energy systems ia a key domain where data-driven methods can significantly improve decision-making. In NYC, buildings account for a large share of total energy consumption and emissions, making them a high-impact target for optimization, so improving how energy-related decisions are made at the building level can lead to substantial reductions in both operational costs and environmental impact. To address this challenge, this project adopts a data-driven approach using machine learning to enable building-level prioritization.

The central decision problem addressed in this project is: How can New York City prioritize buildings for energy efficiency retrofits under limited budget constraints?

Retrofit interventions, such as upgrading HVAC systems, improving insulation, or optimizing lighting, require significant financial investment and cannot be implemented across all buildings simultaneously. Therefore, decision-makers must determine which buildings will yield the greatest benefit if retrofitted first. However, current approaches often rely on broad benchmarks or generalized policies rather than building-level insights. This leads to inefficient investment decisions, where some high-impact buildings may be overlooked while resources are allocated to buildings with lower potential for improvement, ultimately resulting in higher long-term costs and emissions.

This problem is important because NYC represents a high-impact setting where better decision-making can produce large-scale benefits. Given the sheer number of buildings and the scale of energy consumption, even small improvements in prioritization can lead to significant cumulative reductions in both energy usage and emissions. In addition, inefficient buildings impose ongoing financial burdens on owners and operators, making accurate prioritization both an economic and sustainability concern.

The primary beneficiaries of this work include city planners and policymakers, who are responsible for allocating resources and meeting sustainability targets; building owners and facility managers, who seek to reduce energy costs and improve efficiency; and energy and sustainability organizations, which aim to reduce overall energy demand and environmental impact. By improving the precision of retrofit prioritization, this project supports more effective and impactful decision-making across all of these stakeholders.

The core research questions guiding this work are:

1. Can building characteristics predict energy consumption?
2. Which buildings should be prioritized for retrofits?
3. What factors drive energy inefficiency?
4. How can predictions be translated into actionable rankings?

The goal of this project is to predict building energy consumption and generate a data-driven ranking to prioritize buildings for energy efficiency retrofits under limited budget constraints.

Success is evaluated using three key metrics. First, prediction accuracy is measured using Mean Absolute Error (MAE) to assess how accurately energy consumption can be estimated. Second, classification performance is evaluated using the F1-score, which measures the model's ability to correctly identify high-energy buildings. Finally, ranking quality is assessed using Top-K Precision, which evaluates how effectively the model prioritizes the most inefficient buildings for intervention.

II. Data

This project combines multiple datasets to capture different dimensions of building energy performance. The primary dataset used is the NYC Energy Disclosure (Local Law 84) dataset, which provides detailed building-level energy performance information across New York City. Each observation in the datasetr epresents a single building, making the unit of analysis building-level.

Key variables in this dataset include Energy Use Intensity (EUI), which measures energy consumption per square foot, Energy Star Score (a 1–100 metric of overall efficiency), Energy Efficiency Grade (a categorical performance rating), Gross Floor Area (building size), and Property Type (e.g., office, residential, hospital). These features capture the structural and operational characteristics of buildings. A geographic lookup dataset was also used to provide spatial information such as latitude and longitude, enabling location-based analysis. A climate dataset from NOAA was also included, providing environmental variables such as temperature, precipitation, and seasonal trends (winter and summer averages).

Together, these datasets capture three key dimensions of building performance:

- Structural characteristics (e.g., size, property type)
- Performance metrics (e.g., Energy Star Score, EUI)
- Environmental context (e.g., climate conditions)

The primary regression target is Energy Use Intensity (EUI), which serves as a standardized measure of energy consumption. For classification, a binary label was defined to identify high-energy buildings, where buildings in the top 25% of EUI values are labeled as high-energy. This threshold reflects real-world budget constraints by focusing on the most inefficient buildings.

Prior to modeling, the dataset underwent several preprocessing steps. Duplicate building entries were removed, missing values were handled using median imputation, and extreme outliers were filtered to improve stability. Categorical variables such as property type were converted using one-hot encoding, and additional engineered features were created to better capture building characteristics and energy behavior, including derived metrics such as energy intensity ratios, usage-based indicators, and aggregated climate variables.

The dataset was split into training and testing sets using an 80/20 split, and model performance was evaluated using Mean Absolute Error (MAE), which provides an interpretable measure of prediction accuracy. The training set was used to develop the models described in the next section, while the test set was reserved for evaluating model performance on unseen data.

III. Methods

This project employs a multi-stage machine learning framework that combines regression, classification, ranking, and clustering.

This layered approach enables the system to not only predict energy consumption but also translate those predictions into structured decision outputs for retrofit prioritization. The naïve mean predictor and linear regression serve as baseline models, while Random Forest and Gradient Boosting are used as advanced comparison models.

The first stage of the framework focuses on predicting building energy consumption using regression models, which serves as the foundation for identifying inefficient buildings and supporting downstream classification and prioritization.

Two baseline models were implemented to establish a performance benchmark. The naïve baseline predicts the average energy use across all buildings, providing a simple reference point. Linear regression was also used as a baseline model to capture linear relationships between building features and energy consumption. While linear regression improves upon the naïve model, it is limited in its ability to model complex, nonlinear relationships present in building energy data.

To tackle these issues, advanced ensemble models were implemented. A Random Forest model was used to capture nonlinear relationships and interactions between features by aggregating predictions from multiple decision trees. Additionally, a Gradient Boosting model was applied, which builds trees sequentially, focusing on correcting errors from previous iterations. These models are well-suited for tabular data and can effectively model the complex dependencies that drive building energy usage.

Building on the regression outputs, a classification layer was introduced to identify high-energy buildings. A logistic regression model was used to estimate the probability that a building belongs to the top 25% of energy consumption. This threshold reflects real-world budget constraints by focusing on the most inefficient buildings.

The classification model transforms the continuous prediction problem into a binary decision framework, enabling direct identification of buildings that should be considered for retrofit prioritization. By estimating probabilities rather than making only hard classifications, the model also provides additional flexibility for downstream ranking.

To convert model outputs into actionable decisions, a ranking framework was developed. Buildings are ordered based on predicted energy performance and classification probabilities, allowing stakeholders to prioritize retrofit investments effectively. In this framework, buildings with higher predicted energy consumption and higher probability of being classified as high-energy are assigned higher priority. This creates a clear and interpretable prioritization scheme, where limited resources can be allocated to the buildings with the greatest potential impact. This step is essential for translating model predictions into real-world decision-making.

In addition to prediction and classification, a clustering component was implemented to provide deeper insight into building energy patterns. Clustering is not used for prediction directly, but rather to enhance interpretability and support strategy design. K-Means clustering was applied to group buildings with similar energy usage characteristics. The number of clusters was selected using the elbow method, resulting in three clusters. These clusters represent groups of buildings with similar structural and operational profiles, enabling the identification of patterns in energy inefficiency. This insight supports the development of targeted retrofit strategies, as different groups of buildings may benefit from different types of interventions (e.g., HVAC upgrades for offices versus insulation improvements for residential buildings).

IV. Evaluation

The evaluation framework is designed to assess not only predictive accuracy but also how effectively the model supports decision-making for retrofit prioritization. Model performance was first evaluated on a test set to measure generalization to unseen buildings. This ensures that the reported results reflect how the system would perform in a real deployment setting, where predictions must be made on buildings not previously observed during training.

To improve the robustness of these results, k-fold cross-validation was applied on the training data. In this approach, the dataset is partitioned into multiple folds, and the model is trained and validated repeatedly across different splits. This reduces sensitivity to any single train-test partition and provides a more stable estimate of model performance. For classification tasks, stratified sampling was used within cross-validation to preserve the proportion of high-energy and low-energy buildings across folds, ensuring that performance is not biased by class imbalance. Model performance is evaluated using Mean Absolute Error (MAE) for regression, F1-score for classification, and Top-K Precision for ranking quality.

Evaluation was conducted across three complementary dimensions of the problem:

1. Energy prediction: performance is assessed using MAE to quantify the average prediction error between predicted and actual energy consumption. The emphasis here is on consistency of error across building types and the ability of the model to generalize beyond simple patterns captured by baseline approaches.
2. High-energy identification: evaluation focuses on the trade-off between correctly identifying inefficient buildings and avoiding missed opportunities. In this context, false negatives are more costly than false positives, as they represent missed retrofit opportunities. As a result, model behavior is examined not only through aggregate performance but also through error distribution, ensuring that the model aligns with the conservative prioritization strategy described earlier.
3. Ranking performance: evaluation considers how effectively the model prioritizes the most impactful buildings. Since real-world retrofit programs operate under strict budget constraints, only a subset of buildings can be selected. Therefore, performance is judged based on how well the top-ranked buildings correspond to the most inefficient ones. This directly reflects the practical usefulness of the system.

Rather than optimizing solely for prediction accuracy, the framework emphasizes decision relevance, ensuring that model outputs translate into meaningful prioritization outcomes. The combination of test set evaluation and cross-validation provides confidence that the observed performance is both reliable and generalizable, supporting the use of the model in real-world applications. This multi-metric evaluation ensures that models are not only accurate, but also aligned with the practical goal of prioritizing high-impact retrofit decisions.

V. Results

The results reveal a distinct performance gap between baseline and advanced models. The naïve model produced a high prediction error (MAE ≈ 24.9), while linear regression reduced this error to approximately 12.7, indicating that simple feature relationships provide some predictive value.

However, the most significant improvement came from ensemble methods. Random Forest achieved the lowest error (MAE ≈ 7.6), with Gradient Boosting performing similarly (MAE ≈ 8.4), demonstrating that nonlinear models are substantially better at capturing the complexity of building energy usage.

This improvement suggests that energy consumption is not driven by isolated features, but rather by interactions between characteristics such as building type, size, and operational behavior. As a result, models that can capture these interactions provide much more accurate predictions. Feature importance results provide insight into which factors most strongly influence building energy consumption. The model identifies building-specific characteristics, Energy Use Intensity (EUI), gas fraction, and property type, as the most significant predictors. Moderately important features include building size and usage-related characteristics, such as gross floor area and occupancy-related attributes. These variables contribute to variation in energy consumption but do not dominate predictions in the same way as core operational features. In contrast, climate and location-based features exhibit relatively low importance. This suggests that, within a single city like New York, environmental variation is limited compared to differences in building operation and design. As a result, energy consumption is driven more by internal building characteristics than external conditions. These findings indicate that improving building efficiency requires focusing on operational and structural factors rather than geographic or environmental differences.

In classification tasks, Random Forest and Gradient Boosting achieved near-perfect performance, with F1-scores, precision, and recall values close to 1.0 and AUC-ROC values approaching 0.99. Rather than simply indicating high accuracy, this consistency across metrics suggests that the models are reliably identifying high-energy buildings without heavily favoring one type of error over another.

Error patterns provide additional insight into model limitations. While overall accuracy exceeds 90% across building types, misclassifications are concentrated in more complex categories such as offices and hospitals. These buildings exhibit more variable and heterogeneous energy usage patterns, making them inherently more difficult to model. Additionally, the model shows a tendency toward false positives, slightly over-identifying high-energy buildings. This conservative bias is beneficial in the context of retrofit prioritization, as it reduces the risk of missing buildings that would have the greatest impact if improved.

The prioritization framework further demonstrates that model outputs can be translated into a continuous ranking, enabling flexible selection of buildings based on budget constraints. Overall, these results indicate that the value of the framework lies not only in predictive accuracy, but in its ability to produce structured outputs that directly support prioritization decisions.

VI. Interpretation for the System

The results of this project can be directly translated into a practical decision-support workflow for stakeholders responsible for building energy management and retrofit planning. Rather than relying on broad benchmarks or uniform policies, stakeholders can use the model outputs to prioritize buildings based on predicted impact, enabling more efficient allocation of limited resources.

Buildings should be ordered using the model's predicted priority scores, by adopting a rank-based prioritization strategy, and stakeholders can then select the top fraction of buildings that aligns with available budgets or program capacity. For example, if funding is sufficient to retrofit only 10–20% of buildings, decision-makers can focus on the highest-ranked subset, ensuring that investments are directed toward the most energy-intensive and impactful candidates. This approach is flexible and scalable, allowing thresholds to be adjusted as constraints change.

In addition to prioritization, the results support the development of targeted retrofit strategies. Since the model reveals that energy consumption is largely dictated by building-specific characteristics, stakeholders can tailor interventions based on building type and operational patterns. For instance, energy-intensive commercial buildings may benefit most from HVAC system upgrades and control optimization, while residential buildings may require insulation improvements or appliance efficiency upgrades. Using clustering insights alongside ranking allows stakeholders to group similar buildings and apply standardized retrofit packages, improving both efficiency and cost-effectiveness.

The framework also enables proactive decision-making. Instead of reacting to high energy usage after it occurs, stakeholders can use predictions to identify buildings that are likely to be inefficient and intervene earlier. This shifts energy management from a reactive process to a more strategic and preventative one, improving long-term outcomes.

An important consideration is the model's tendency toward conservative classification, resulting in some buildings being over-identified as high-energy. While this approach minimizes the risk of missing critical retrofit opportunities, it may lead to increased evaluation costs for buildings that are not ultimately high priority. As a result, model outputs should be used as a decision-support tool and supplemented with domain expertise and on-site validation prior to final investment decisions.

Finally, the framework assumes relatively stable environmental conditions within the city, which may limit its ability to generalize to regions with more significant climate variation. When applying this approach to other locations, additional calibration may be required to account for regional differences. This ensures that the system remains both actionable and adaptable across different policy and budget scenarios.

VII. Limitations and Next Steps

While the proposed framework demonstrates strong predictive performance and decision-making potential, several limitations must be considered. The model relies on historical data, assuming that past energy usage patterns will persist over time. However, changes in occupancy behavior, building operations, or policy regulations could reduce prediction accuracy. In addition, the prioritization framework does not account for feasibility constraints, meaning that some high-priority buildings may not be practical to retrofit due to financial, structural, or regulatory limitations. As a result, the model should be used as a decision-support tool rather than a standalone decision rule.

A key data limitation is the use of annual energy consumption values, which prevents the model from capturing seasonal and short-term variation. In reality, energy demand fluctuates due to weather conditions, occupancy patterns, and operational schedules. Aggregating data at the annual level may obscure important dynamics such as peak usage periods and time-specific inefficiencies, limiting the model's ability to fully represent real-world behavior. This issue is compounded by the limited variability in climate features, as all buildings are linked to the same NYC (Central Park) climate data. As a result, environmental factors do not meaningfully differentiate buildings, which may lead the model to underestimate their impact and limit its ability to generalize to regions with more diverse climates.

Another important limitation is the lack of detailed building-level characteristics. Key factors such as building age, HVAC system type, insulation quality, and occupancy behavior are not included in the dataset. These variables are critical drivers of energy efficiency, and without them, it is not possible to determine which specific retrofit strategies (HVAC upgrades, insulation improvements, or lighting replacements) are most appropriate for each building. This limits the ability to translate model outputs into targeted, actionable interventions.

Incorporating higher-resolution, time-series energy data (e.g., monthly or hourly measurements) would allow the model to capture seasonal trends and operational variability, improving prediction accuracy and prioritization. Integrating cost and financial data, such as retrofit implementation costs, expected energy savings, maintenance costs, and payback periods, would enable a return-on-investment (ROI)-based prioritization framework. This would shift the system from identifying high-energy buildings to identifying those that provide the greatest economic and environmental return per dollar invested. Additionally, expanding the dataset to include more detailed building characteristics would improve both predictive accuracy and interpretability. Overall, the effectiveness of the framework depends more on improving data quality and incorporating decision-relevant features than on increasing model complexity.